

Performance Testing

Date	29 September 2026
Team ID	6ab4ce760fc666a751b52130
Project Name	AI StudyBuddy
Maximum Marks	5 Marks

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman
Type of Testing	Functional Testing, API Response Testing, Load Testing (manual)
Target Module / API	All 18 REST API endpoints — Auth, Materials, AI, Admin
Test Environment	Local development server (localhost:5000) + MongoDB Atlas (cloud)
Test Date	28 September 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	User Registration — POST /api/auth/register	1	< 2	201 Created with JWT tokens and user info
2	User Login — POST /api/auth/login	1	< 2	200 OK with JWT access and refresh tokens
3	Upload Study Material — POST /api/materials/upload	1	< 3	201 Created with material record (content extracted)
4	Generate AI Summary — POST /api/materials/:id/summarize	1	< 10	200 OK with concise bullet-point summary from Gemini
5	Generate AI Flashcards — POST /api/materials/:id/flashcards	1	< 10	200 OK with JSON array of Q&A flashcards
6	Generate AI Quiz — POST /api/materials/:id/quiz	1	< 10	200 OK with JSON array of MCQ questions
7	Generate AI Study Plan — POST /api/materials/:id/study-plan	1	< 10	200 OK with personalized day-by-day study schedule
8	Access Protected Route Without Token	1	< 1	401 Not authenticated
9	Upload Invalid File Type (.jpg)	1	< 1	400 Error — Only .txt, .md, and .pdf files allowed
10	Admin Get Platform Stats — GET /api/admin/stats	1	< 2	200 OK with totalUsers and totalMaterials count

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg) — Auth endpoints	< 2 seconds	~500ms	Pass	Registration and login respond quickly
2	Response Time (Max) — AI endpoints	< 10 seconds	~3-8 seconds	Pass	AI generation time depends on Gemini API and material length
3	Throughput (Req/sec) — CRUD endpoints	N/A	Adequate for single-user testing	Pass	Performs well for expected user load
4	Error Rate	< 1%	0%	Pass	All valid requests processed successfully
5	CPU Utilization	< 80%	Low (< 30%)	Pass	Node.js event loop handles async operations efficiently
6	Memory Utilization	< 80%	Low (< 40%)	Pass	Temp files cleaned up after text extraction

Step 4: Observations & Analysis

- All 18 API endpoints return correct HTTP status codes and expected JSON responses.
- Auth endpoints (register, login, refresh, logout) respond within 500ms.
- Material upload extracts content successfully from .txt, .md, and .pdf files and cleans up temp files.
- AI generation endpoints (summarize, flashcards, quiz, study-plan) take 3-8 seconds depending on material length and Gemini API response time.
- Error handling works correctly — invalid tokens return 401, invalid file types return 400, unauthorized admin access returns 403.
- All AI-generated content is correctly persisted to MongoDB and retrievable via GET endpoints.
- The express-async-errors package catches all async errors without crashing the server.

Step 5: Screenshots / Evidence

(Attach Postman screenshots showing API test results for each endpoint — registration response, login response, upload response, AI summary response, flashcard JSON response, quiz JSON response, study plan response, error responses)